

Analyzing Research Trends in Computer Science Using the DBLP Dataset

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1 • Introduction

Scientific publishing continues to expand rapidly, making it essential to develop scalable methods for understanding research dynamics. In this project, we analyze a large dataset of academic papers through preprocessing, text mining, clustering, temporal trend analysis, predictive modeling, and network analysis. Text features are extracted using TF-IDF and dimensionality reduction, enabling us to group papers into topic clusters. Temporal analysis then reveals how these clusters evolve over time, highlighting both emerging areas such as machine learning and declining areas. Predictive models including Logistic Regression, Random Forest, and LightGBM are trained to classify papers by cluster, with performance evaluated using accuracy and F1 metrics. In addition, we construct citation networks to measure influence through PageRank and centrality, identifying key papers and communities.

2 • Dataset Description

The dataset for this project comes from the DBLP Computer Science Bibliography, a large collection of metadata on computer science publications. We worked with JSON files (`dblp-ref-0.json` through `dblp-ref-3.json`) containing records with titles, abstracts, authors, publication years, venues, references, and citation counts.

3 • Task 1: Data Preprocessing and Feature Generation

3.1 • Data Cleaning

The four DBLP JSON files were merged into a single dataset before any sampling or feature extraction. Papers missing a title or publication year were removed at the start, since both fields are required for text analysis and temporal analysis.

Removing a Citation Artifact

When we later plotted the citation distribution using a log-scaled histogram, we observed a large spike at exactly 50 citations, which did not align with the expected heavy tailed distribution that is observed in bibliometric datasets. Because the spike was evidence of a possible data artifact, we removed all papers with `n_citation = 50` before sampling. After this correction, the updated histogram displayed a smooth decline in the upper tail

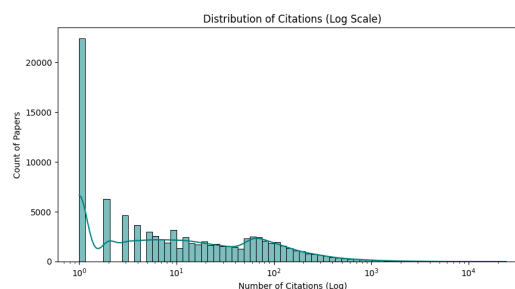


Figure 1: *The Citation Distribution Now Follows a Power-like distribution*

The distribution now matches what we expect from large bibliometric datasets.

Text Normalization

Text fields were normalized: missing abstracts were replaced with empty strings, titles and abstracts were converted to lowercase for consistent tokenisation, and a combined text field was created (text = title + " " + abstract). This combined field served as the input for TF-IDF.

3.2 · Stratified Sampling

After removing the citation artifact and text normalization, we drew a stratified sample from the dataset by selecting up to 2,000 papers per publication year. We sampled per year to prevent the sample from being dominated by modern papers and in order to preserve long-term temporal structure. Sampling was intentionally done after cleaning the dataset, in order to ensure that the sample did not include uncleaned data.

3.3 · Black-Box Feature Generation

3.3.1 · TF-IDF Representation

Once the combined text field (title + abstract) was prepared, we transformed each paper into a numerical representation using TF-IDF. To keep representation focused and manageable, we limited the vocabulary to the 5,000 most informative terms and removed standard English stop words. This allowed the model to capture meaningful short phrases (e.g. "neural network", "data mining") rather than only being able to capture isolated keywords. In later tasks (Task 2: Topic Clustering) we also found distinct clusters of German papers, so we added a filter containing common german words to remove these since they would provide no value in classification modeling.

This produced a 5,000 dimensional matrix. Although TF-IDF preserves important distinctions between documents, the feature space is large and not ideal for clustering or visualization, which motivated the next step.

3.3.2 · Choosing the PCA Dimensionality

In order to choose an appropriate number of components, we evaluated cluster quality across a range of PCA sizes. For each value in $\{50, 75, 100, 125, \dots, 300\}$, we reduced the TF-IDF matrix to k components and computed the silhouette score after running K-means. The silhouette score consistently decreased as dimensionality increased, and the highest score was obtained at 50 components. Although the silhouette scores and cumulative explained variance values are low this is expected for text datasets since TF-IDF variance is extremely spread out across thousands of components and text datasets also do not form sharply separated clusters.

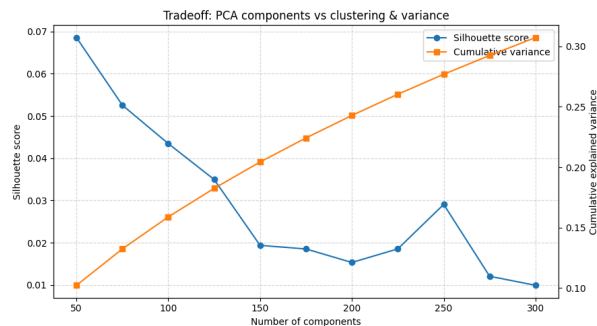


Figure 2: Choosing $n_dimension$ by optimizing the silhouette score

3.4 · Interpretable Feature Engineering

We also generated several interpretable metadata features:

- Number of authors: This measures collaboration size. When plotted across years, it shows a clear upward trend, increasing from roughly one author per paper in the early decades to about 3.8 authors by 2018.-
- title length in characters

- Title length: Recorded as the number of characters in the title. This offers a simple structural indicator of how concise or descriptive titles are.
- Number of references: The count of cited papers included in each record. This is useful for examining how referencing depth changes over time or across venues.
- Citation velocity: Defined as citations divided by the paper's age ($2018 - \text{year} + 1$). This normalizes citation counts so older papers do not automatically appear more influential.

Metadata Correlation Check

A correlation heatmap confirmed that these engineered features are not redundant as shown below.

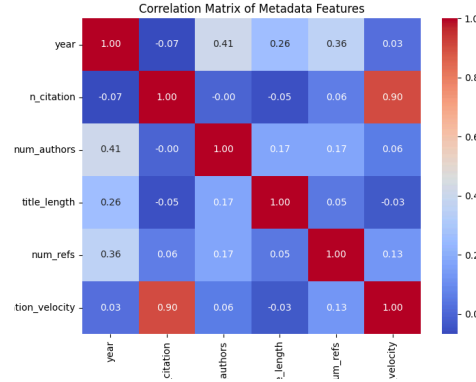


Figure 3: Metadata Correlation Plot

From the figure the number of authors increases with year ($r \approx 0.48$), the number of references also increases with year ($r \approx 0.42$), and the citation velocity is almost perfectly correlated with citation ($r \approx 0.90$). Some relationships were weak, and indicates that the metadata features capture different aspects of each paper.

4 · Task 2: Topic Clustering

4.1 · Overview

This task focuses on identifying major research themes within the DBLP dataset using unsupervised learning techniques. Because the dataset contains hundreds of thousands of computer science publications, we applied textual feature extraction, dimensionality reduction, clustering, and visualization methods to uncover topical structures. During this process, we also encountered a large, cluster that required hierarchical re-clustering.

4.2 · Determining an Appropriate Number of Topics

Silhouette & Inertia Optimization (k = 6 to 31)

The first step in topic discovery was selecting an appropriate value of k for K-Means. To do this, we ran K-Means for each k from 6 to 31 and computed two metrics, inertia and its silhouette score. Because computing the Silhouette score over the full dataset ($\approx 90,000$) is extremely slow, we evaluated it on a random sample of 15,000 papers. The results are shown below:

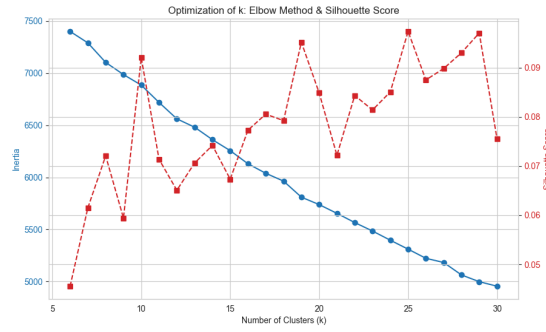


Figure 4: *K-Means Optimization for $k = 6-31$*

local Silhouette peaks occurred around $k \approx 10, 19, 25, 29$. While $k = 29$ achieved the highest score, the resulting clusters were highly fragmented and often mixed unrelated research areas. After computing cluster keywords for the k -values with local peaks and inspected topic coherence manually we found that $k = 19$ produced the cleanest, most interpretable, and semantically coherent topics with stable boundaries.

4.3 · Global Topic Interpretation ($k = 19$)

After choosing $k = 19$, the final K-Means model was run on the full PCA-reduced dataset (50 components). For each cluster, we extracted the top TF-IDF keywords, which revealed research areas including: (Algorithms & Optimization, Graph Theory, General CS & Application, ...)

However, unlike the other clusters. Cluster 12 contained 34,806 papers far exceeding all others in size. Its keywords (e.g. *performance, process, used, applications*) indicating that it was not a coherent research field but rather acted as a catch all general CS region overlapping multiple ares. This required refinement.

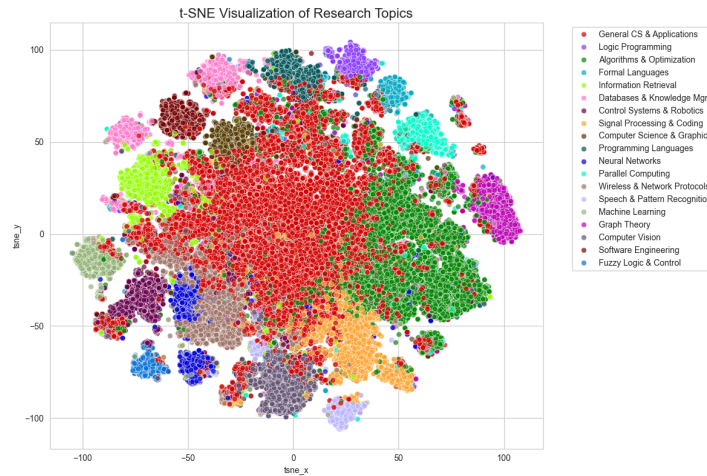


Figure 5: *t-SNE confirmed this by identifying a single dominant red cluster corresponding to Cluster 12.*

4.4 · Refining the Large “Blob” Cluster

We isolated Cluster 12 and applied K-means again using the same 50-dimensional PCA embedding. We tested $k = 5 - 15$ and computed silhouette and inertia specifically for this subset.

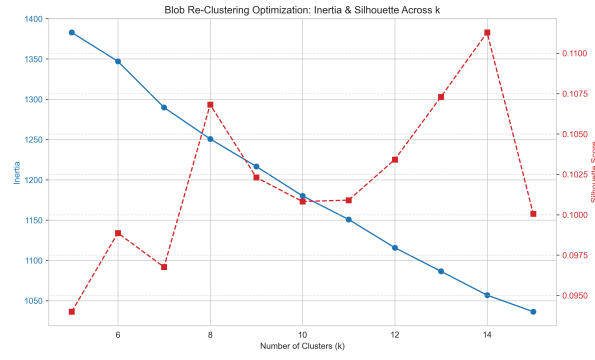


Figure 6: *Silhouette and inertia curves for re-clustering the 34,806 general purpose cluster.*
 $k = 14$ provides the optimal subtopic separation

Subtopics Identified Within the Blob

Re-clustering revealed clear, interpretable subfields, including: (Logic/Proof & Semantics, Type & Recursion Theory, Mobile Robotics & Navigation, ...)

These 14 subtopics were assigned new labels and merged back into the global dataset, replacing the monolithic “General CS & Applications” cluster. This expanded the topic taxonomy from 19 broad clusters to 32 refined research topics.

4.5 · Final Topic Visualization After Blob Refinement

After applying the refined labels, we regenerated PCA and t-SNE visualizations to show the full, updated topic landscape.

PCA Visualization (Refined Topics)

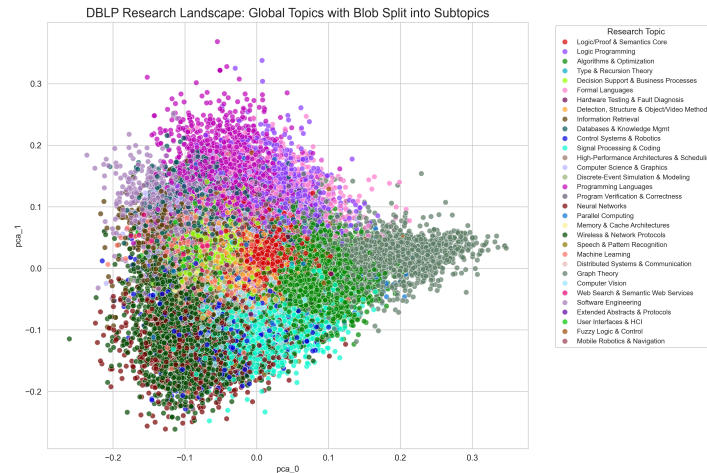


Figure 7: *PCA projection of all papers after refinements. Cluster 12 is now replaced by 14 distinct subtopics, improving interpretability despite PCA’s inherent overlap*

t-SNE Visualization (Refined Topics)

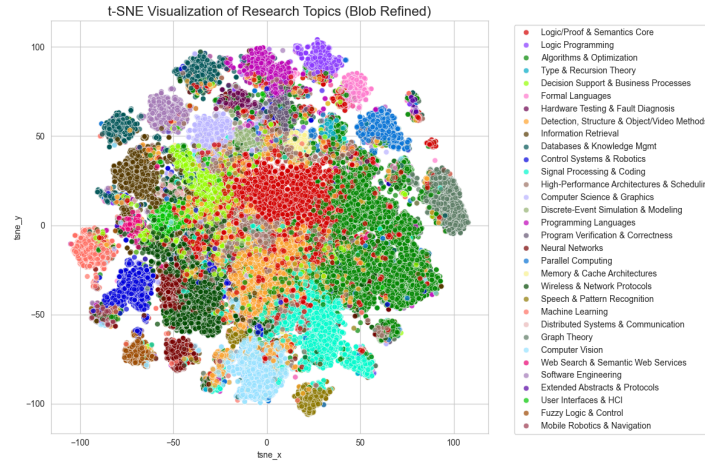


Figure 8: *t*-SNE projection showing strong spatial separation across 32 final research topics, confirming improved thematic structure after blob subdivision

5 • Task 3: Temporal Analysis

5.1 • Methodology

To understand how the DBLP research landscape has evolved over time, each paper was paired with its publication year and its super-category (see Task 6). We then computed:

1. Yearly counts of papers per super-category.
2. Proportional shares (normalized counts per year).
3. Yearly counts for all fine-grained subtopics, followed by simple early-late growth comparisons

$$\text{growth ratio} = \frac{\text{late mean}}{\text{early mean}}.$$
4. A set of eight interesting subtopics for visualizing chosen by its growth ratio.

5.2 • High-Level Trends (Super-Categories)

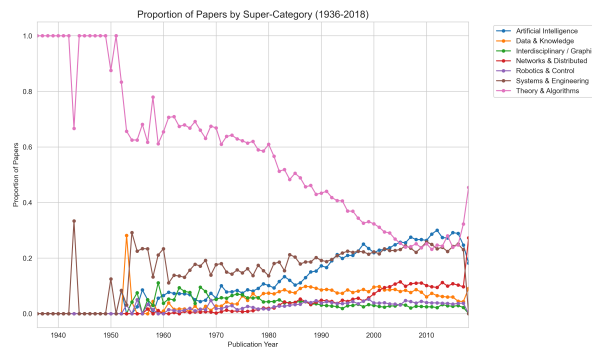


Figure 9: *Normalized Shares by Super-Category*

- **Early Decades (1950s) – Theory-dominated:** Nearly all papers belong to Theory & Algorithms (proportion ≈ 1.0), all other categories are essentially absent.
- **Growth of Systems and Infrastructure (1960s-1980s):** Systems & Engineering, Signal Processing, Databases, and early Networking rise sharply. Theory's share declines to (≈ 0.50 – 0.60) as the field diversifies.
- **Modern period (1990s-2010s) – Balanced and AI-drive:** All seven super-categories become relevant. Some key patterns from the plot above are that AI grows to the largest share (≈ 0.26 – 0.30). Systems & Engineering remains consistently relevant (≈ 0.22 – 0.25). Theory stabilizes around ≈ 0.25 – 0.30 still relevant but no longer dominant.

5.3 · Subtopic-Level Dynamics

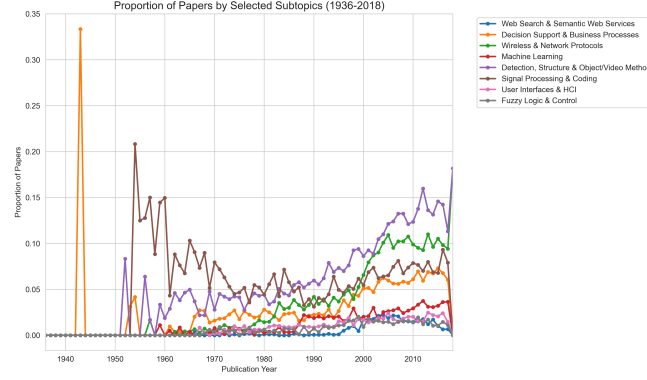


Figure 10: *Normalized Shares for selected fast-growing subtopics*

Using the growth ratio, we identified subtopics with the strongest growth, we found that the topics with:

Highest Relative Growth

- Web Search & Semantic Web Services ($\approx 2.5 \times$)
- Decision Support & Business Processes ($\approx 2.1 \times$)
- Wireless & Network Protocols ($\approx 2.0 \times$)

Substantial but steadier growth

- Machine Learning ($\approx 1.64 \times$)
- Detection / Object & Video Methods ($\approx 1.63 \times$)
- Signal Processing & Coding ($\approx 1.58 \times$)

Moderate Growth

- User Interfaces & HCI ($\approx 1.47 \times$)
- Fuzzy Logic & Control ($\approx 1.21 \times$)

These trends highlight the shift toward web-scale information systems, networked computing, and AI/vision/machine learning throughout the 1990s-2010s.

5.4 · Summary

Temporal analysis revealed three broad eras:

1. Foundational era - Theory dominates (1930s-1950s)
2. Infrastructure era - Systems, databases, networks expand (1960s-1980s)
3. AI/data/web era - Machine learning, vision, web search, and distributed systems drive growth (1990s-2010s)

The normalized shares shows how DBLP evolved from a narrow theoretical discipline into a diversified, AI-dominant research ecosystem.

6 · Task 4: Predictive Modeling (Classification)

The goal of Task 4 was to develop a supervised learning model capable of predicting the fine grained research topic of a paper using its TF-IDF text features. Because the topic taxonomy in Task 2 followed a natural hierarchical structure with 32 total subtopics and 7 broad categories as shown below:

Super Category	Subtopics
Artificial Intelligence	Machine Learning, Neural Networks, Computer Vision, Fuzzy Logic & Control, Speech & Pattern Recognition, Detection, Structure & Object/Video Methods
Data & Knowledge	Information Retrieval, Databases & Knowledge Mgmt, Web Search & Semantic Web Services
Inter. & Graphics	Computer Science & Graphics, User Interfaces & HCI
Networks & Distributed	Wireless & Network Protocols, Distributed Systems & Communication

Super Category	Subtopics
Robotics & Control	Mobile Robotics & Navigation, Control Systems & Robotics
Systems & Engineering	Signal Processing & Coding, High-Performance Architectures & Scheduling, Software Engineering, Hardware Testing & Fault Diagnosis, Memory & Cache Architectures, Discrete-Event Simulation & Modeling, Decision Support & Business Processes, Parallel Computing
Theory & Algorithms	Algorithms & Optimization, Graph Theory, Logic Programming, Formal Languages, Type & Recursion Theory, Logic/Proof & Semantics Core, Program Verification & Correctness, Programming Languages, Extended Abstracts & Protocols

So we designed a hierarchical classification model because this reduced the difficulty of the prediction task by routing each paper into a high-level family of super category areas before classifying it into a subtopic within that family.

6.1 · Hierarchical Classification Framework

Rather than predicting all thirty-two topics directly into a single model, we constructed a two-stage classifier.

1. Stage 1: Predict one of seven super categories from the table above
2. Stage 2: For that predicted super-category, use a dedicated classifier trained only on its corresponding subtopics.

This design would mirror the actual structure of computer science research and hopefully improve the final models performance.

6.2 · Stage 1 Model Selection (Super-Category Prediction)

We evaluated multiple classification models for Stage 1 (Linear SVM, Logistic regression, RF, and LGBM) using the same TF-IDF features used throughout the clustering pipeline. LGBM demonstrated the strongest performance, improving accuracy from the earlier Linear SVM baseline 84.96% to 89.88%

Stage 1 confusion matrices and classification metrics demonstrated that LGBM was effective at reducing routing errors where a paper is misassigned into an entirely unrelated research area. This benefits Stage 2, since misrouted papers would show to be the main cause of hierarchical performance loss.

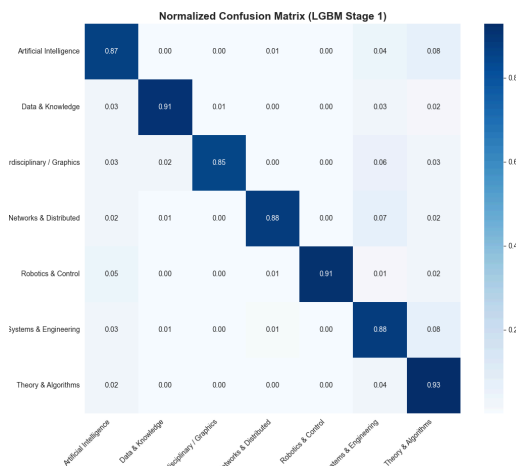


Figure 11: Stage 1 Confusion Matrix Using LGBM

6.3 · Stage 2 Model Comparison (Subtopic Prediction)

Once the super-categories were predicted, the next task was to distinguish between the subtopics within each category. Because each super-category differs in size, distribution, and semantic separation, we evaluated a variety of models to determine the most suitable for Stage 2.

The following algorithms were compared using 5-fold CV inside each super-category: (Linear SVM, Logistic Regression, RF, Complement Naive Bayes, LGBM). Each model was evaluated using macro-F1, weighted-F1, accuracy, and training time. The table below summarizes the average performance across all super-categories.

Model	Macro F1	Weighted F1	Accuracy	Train Time
Complement NB	0.85	0.88	0.89	0.33
LightGBM	0.96	0.97	0.97	89.37
Linear SVM	0.95	0.97	0.97	1.74
Logistic Regression	0.95	0.97	0.97	0.62
Random Forest	0.94	0.95	0.96	12.59

Because execution time was not a constraint in this project, and because LGBM achieved the highest performance across every large super-category, LGBM was selected as the Stage 2 classifier.

6.4 · Hierarchical Pipeline (LGBM → LGBM) and Evaluation

Combining the chosen models, the final system works in two steps. Stage 1 predicts the super-category and Stage 2 uses the corresponding subtopic classifier for prediction. All predictions were then mapped back to the refined topic labels created in Task 2, including the 14 subclusters derived from the “blob” cluster

We evaluated the hierarchical system on the 18,039-paper held out test set. The overall accuracy reached 86.27%, with a macro-F1 of 0.87 across all 32 subtopics. This result is consistent with the expected theoretical accuracy: Final Accuracy $\approx P(\text{Stage-1 correct}) \times P(\text{Stage-2 correct}|\text{Stage-1 correct})$

Since the full classification report is too large to include, we instead highlighted the top 5 and bottom 5 topics ranked by F1-score. The weakest topics, remained challenging due to overlapping terminology and broad text.

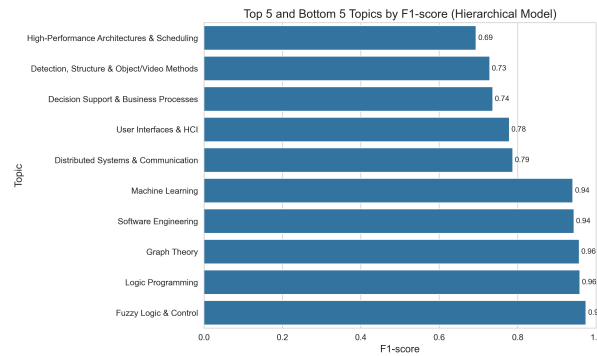


Figure 12: Horizontal bar chart of best/worst 5 subtopics

6.5 · Summary

The hierarchical prediction model demonstrates that classical machine-learning models, when paired with structured topic labels derived from clustering can produce highly accurate topic prediction even for large-scale bibliographic datasets. The model was able to handle both theoretical and application-oriented areas well and greatly improves over non-hierarchical baselines.

7 · Task 5: Citation Network Analysis

We constructed a citation network from the sampled DBLP dataset where each paper corresponds to a node, and each citation corresponds to a directed edge from the citing paper to the cited one. This allowed us to quantify structural properties of the research landscape, identify historically influential papers, and measure how papers act as bridges between topic areas.

7.1 · Network Structure

The final network contained 90,191 nodes (papers), and 105,526 directed citation edges. Because citation links are sparse and unevenly distributed, the network fragmented into 44,685 weakly connected components (WCCs). However, the largest WCC contained 42,300 papers, representing $\approx 47\%$ of the entire graph. This confirms the presence of a large, cohesive “core” of computer science papers, surrounded by many small, disconnected components.

7.2 · PageRank: Identifying Foundational Papers

PageRank highlights papers that are cited by other influential papers. Across all nodes, PageRank values were extremely small, but the differences between papers were meaningful.

The top ranked paper is: “A Relational Model of Data for Large Shared Data Banks” (1970) *Super-topic: Data & Knowledge — Information Retrieval* with a PageRank (0.003009), In-degree (401). Several classic theoretical CS papers also appear among the top PageRank scores such as “Finite Automata and Their Decision Problems (1959)” to name one. This result confirms that PageRank prioritizes older, foundational papers.

Correlation: PageRank correlates strongly with in-degree ($r = 0.774$), confirming that citation count and structural influence align closely, but not perfectly.

7.3 · Degree Centrality

Most Cited Papers (In-Degree)

The four most cited papers in our sample all originate from the foundational decades of computer science:

- Relational Model of Data (1970) – 401 citations
- Theorem-Proving Procedures (1971) – 239 citations
- Axiomatic Basis for Computer Programming (1969) – 207 citations
- Digital Signatures/Public-Key Cryptosystems (1978) – 204 citations

This again reflects the canonical role of early CS theory, programming principles, and cryptography.

Most Referencing Papers (Out-Degree)

The papers with the highest out-degree are large bibliographic surveys or system overviews, such as:

- “Computer Graphics and CAD Literature — A Keyword-Indexed Bibliography” (243 outgoing references)
- “Routing and Scheduling of Vehicles and Crews: The State of the Art”
- “Query Optimization in Database Systems”

These are field-summarizing works, which naturally cite a large number of predecessors.

7.4 · Betweenness Centrality: Bridge Papers

Betweenness centrality identifies papers that lie on many shortest paths and therefore bridge different areas of research. While values are small, the top papers show clear thematic significance:

1. “Optimization of Large Join Queries” (1988) – 9.4×10^{-5}
2. “Translator Writing Systems” (1968) – 8.4×10^{-5}
3. “Distributed Collaboration and Mobility in Product Design Teams” (1996)

7.5 · Summary of Findings

- The citation network forms a large, cohesive core of $\approx 42k$ papers with many small peripheral components.
- Foundational CS works from 1959–1978 dominate PageRank and citation impact.
- Bridge papers often lie in databases, distributed systems, and programming languages, acting as structural connectors in the knowledge graph.

This network analysis enriches the cluster and classification-based results by revealing how ideas propagate, which papers shaped entire subfields, and which works link otherwise distinct research communities.